Future-Stable Readout Monitors for Independent Write-Once Sources: Exact State Bounds, Matroid-Rank Structure, and Constrained Projection Extremals

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Abstract

Independent write-once binary sources admit a deterministic readout interface that preserves action enabledness and every future protected output. We establish an exact formula for its minimum number of states, expressed through coordinate projections and an incomplete-class correction. For marginally surjective protected label families, we determine sharp thresholds at cardinalities two, three and four, and classify all four-label minimizers. We show that the complete cube-face dimension vector does not determine readout complexity. Finally, the protected-set cost admits a decomposition into truncated partition-matroid rank functions, yielding submodularity and an exact marginal-cost formula for adding protected labels.

Keywords: deterministic readout monitors; write-once sources; state complexity; Boolean projections; matroid ranks; submodular functions.
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1 Introduction

Monitoring concurrent sources requires more than recording their currently observable outputs: future assignments can distinguish source histories that presently look identical. The state complexity of deterministic interfaces is governed by future equivalence [9], while the combinatorics of Boolean projections and cube faces has an independent extremal theory [1, 2, 3]. We bring these ingredients together for a readout port that preserves the enabledness of every source assignment and recognizes designated complete labels.

The problem is formulated in a two-component architecture: an exact source layer retains its semantic configuration, and an autonomous, separately budgeted readout monitor provides protected observations. The resulting minimum port-state count is a concrete invariant with an exact projection formula. Work on semantic normalization and preservation of stack information provides broader representation-theoretic context [10, 11].

Main contributions

(i) An exact minimum-state index for future-stable readout ports that preserve both outputs and source-action enabledness.

(ii) Sharp minimum port budgets for two, three, and four protected complete labels in every source dimension, including the rigidity of four-label optimizers.

(iii) An all-dimensional pair of protected families with the same cube-face dimension vector but different readout-state complexity.

(iv) A truncated partition-matroid rank decomposition, submodularity, and an exact incremental cost formula for protected labels.

2 The physical readout-port contract

Fix $d \geq 2$, write $[d] = \{1, \ldots, d\}$, and let $M = \{\bot, 0, 1\}^d$. The root is $\bot^d$. For each $i \in [d]$ and $a \in \{0, 1\}$, $X_{i,a}$ is defined precisely when coordinate i is unassigned and sets it to a. These actions are independent whenever both are enabled on distinct coordinates.

An uncharged exact source layer retains the full state $s \in M$. A separately charged readout port is a deterministic partial-action Moore machine $(H, h_0, (V_{i,a}), o)$ with a history-independent factor map $h : M \to H$ that intertwines the actions, reflects their enabledness, and outputs a distinct protected label k at the complete state $k \in K \subseteq \{0, 1\}^d$, with a generic output elsewhere. The port alone, not the exact source layer, answers protected observations. Its cost is $|H|$.

For $U \subseteq [d]$ let $\pi_U K$ be the set of restrictions of elements of K to U, and let $A_U = \{0, 1\}^U$. A protected set is *marginally surjective* if each coordinate assumes both values among its elements.

Theorem 2.1 (Exact autonomous-port index)**.** *For any nonempty $K \subseteq \{0, 1\}^d$, the least possible port-state count in the preceding category is*

$$I_d(K) = \sum_{U \subseteq [d]} \left(|\pi_U K| + \mathbf{1}_{\{\pi_U K \neq A_U\}} \right). \tag{1}$$

Proof. A partially assigned source state has a uniquely determined assigned-coordinate support U. Enabledness reflection prevents identification of states with distinct supports: an action on a coordinate in their symmetric difference is enabled at exactly one of them. Fix U. Every partial assignment in $\pi_U K$ has a protected completion; two distinct such assignments are distinguished by completing to one of their protected final labels, which is impossible from the other. A partial assignment outside $\pi_U K$ has no protected completion, and all such assignments have identical future protected outputs and enabledness, so they may be merged into one dead residual class for that support. These classes are stable under the action maps: extensions of a protected prefix are protected prefixes or become dead, and extensions of a dead prefix remain dead. Taking precisely these classes gives a well-defined deterministic port attaining the stated count. □

Remark 2.2 (Interpretation of the port cost)**.** *The port may have fewer states than M because it does not replace the exact source layer. In particular, $I_d(K)$ is not the minimum total physical state count of the composed implementation. This distinction is essential for interpreting any positive assembly defect.*

3 Extremal protected budgets

For $2 \le k \le 2^d$ let

$$f_d(k) = \min\{I_d(K) : K \subseteq \{0,1\}^d,\ |K| = k,\ K \text{ marginally surjective}\}.$$

For budgets admitting such K, define $\mathrm{Sat}_d(b) = \max\{k : f_d(k) \le b\}$. The function Sat_d measures the jointly protectable label population in the stated readout-port architecture.

Theorem 3.1 (First two thresholds)**.** *For every $d \ge 2$,*

$$f_d(2) = 3 \cdot 2^d - d - 2, \qquad f_d(3) = 7 \cdot 2^{d-1} - d - 3.$$

Proof. A marginally surjective two-element set must consist of complementary words. A singleton projection has size two and costs two, and every support of size at least two projects to two words but omits an ambient assignment, costing three. Summing gives the first equality.

For three distinct rows, every nonconstant coordinate partitions the rows into a singleton versus a pair. Let $r \in \{2,3\}$ be the number of occurring partition types, with multiplicities $n_1, \dots, n_r \ge 1$ summing to d. For a support of size at least two, the port contribution is three exactly when all its coordinates have one type and four otherwise; the empty-support contribution is one and singleton contributions are two. Thus

$$I_d(K) = 4 \cdot 2^d - 3 - d + r - \sum_{i=1}^{r} 2^{n_i}.$$

By strict convexity, the maximum of $\sum_i (2^{n_i} - 1)$ subject to $r \ge 2$ occurs for $r = 2$ with multiplicities $(d-1, 1)$. This gives the lower bound, attained by $\{0^d, 1^{d-1}0, 1^{d-1}1\}$. □

Theorem 3.2 (Four-label threshold and rigidity)**.** *For every $d \ge 2$,*

$$f_d(4) = 4 \cdot 2^d - 2d - 3.$$

Equality holds precisely for the protected sets equivalent, up to row relabeling and independent coordinate permutations and bit-complements, to $\{0^{d-1}, 1^{d-1}\} \times \{0,1\}$.

Proof. Label the four protected rows and identify each nonconstant coordinate with its unordered nontrivial bipartition (cut type) of these rows. Let $\mathcal{M}$ contain supports of size at least two on which all coordinates have the same cut type. Let $\mathcal{F}$ contain supports of size at least three that distinguish all four rows. Directly comparing the contribution of each support in (1) with a baseline value four yields

$$I_d(K) = 4 \cdot 2^d - 2d - 3 + |\mathcal{F}| - |\mathcal{M}|.$$

For each occurring cut type q choose one representative coordinate r_q. Map a monochromatic support $U \in \mathcal{M}$ of type p to

$$\Phi(U) = U \cup \{r_q : q \ne p\}.$$

Since the full family of cut types distinguishes all four rows, $\Phi(U) \in \mathcal{F}$. Type p is the unique type repeated in $\Phi(U)$; all added types occur once. This recovers p and then U, so Φ is injective, proving $|\mathcal{F}| \ge |\mathcal{M}|$.

If at least three types occur, the set of the chosen representatives belongs to $\mathcal{F}$ but not to the image of Φ, giving strict inequality. With exactly two types, distinction of all four rows forces both cuts to be balanced and transversal. If each type occurs at least twice, a support with two coordinates of each type lies in $\mathcal{F}$ but outside the image. Thus equality forces multiplicities $(d-1, 1)$. Conversely, in this case every support in $\mathcal{F}$ is obtained by adjoining the unique singleton-type coordinate to a monochromatic support of the repeated type, and the injection is onto. □

4 Cube-face statistics and the missing-class correction

For a protected set K, let $c_j(K)$ count the j-dimensional faces of the d-cube meeting K and let

$$F_d(K) = \sum_{U \subseteq [d]} |\pi_U K| = \sum_{j=0}^{d} c_j(K), \qquad Q_d(K) = \#\{U \subseteq [d] : \pi_U K \neq A_U\}.$$

Then $I_d = F_d + Q_d$. The complete vector $(c_0, \ldots, c_d)$ therefore does not necessarily determine the port complexity.

Proposition 4.1 (Face-vector non-determination in every dimension)**.** *Take* $K_A = \{000, 001, 010, 111\}$ *and* $K_B = \{000, 001, 110, 111\}$. *For each* $d \geq 3$ *define*

$$K_A^{(d)} = K_A \times \{0,1\}^{d-3}, \qquad K_B^{(d)} = K_B \times \{0,1\}^{d-3}.$$

Both sets are marginally surjective, and their entire face-dimension polynomials coincide:

$$\sum_j c_j(K_A^{(d)}) t^j = \sum_j c_j(K_B^{(d)}) t^j = (4 + 10t + 6t^2 + t^3)(2+t)^{d-3}.$$

Nevertheless

$$I_d(K_A^{(d)}) - I_d(K_B^{(d)}) = 2^{d-3} > 0.$$

Proof. A face of a product cube is the product of faces, so its face-intersection polynomial multiplies by $2 + t$ per added free coordinate. In dimension three both sets meet respectively 4, 10, 6, 1 faces of dimensions 0,1,2,3, yielding the claimed common polynomial. The extra coordinate projections of the Cartesian product contribute a factor 3^{d-3} to F and a factor 2^{d-3} to Q. In dimension three, $F_3(K_A) = F_3(K_B) = 21$, $Q_3(K_A) = 3$, and $Q_3(K_B) = 2$, which proves the assertion. □

5 Exact reduction to contained faces and VC shattering

The principal combinatorial quantities admit an exact classical interpretation. Let $C(K^c)$ be the number of all Boolean subcube faces entirely contained in the complement K^c, and let $\mathrm{Shat}(K)$ denote the family of coordinate supports U for which $\pi_U K = \{0,1\}^U$ (the usual shattered sets).

Proposition 5.1 (Classical-invariant decomposition)**.** *For every nonempty protected set,*

$$I_d(K) = 3^d + 2^d - C(K^c) - |\mathrm{Shat}(K)|.$$

Proof. Each Boolean coordinate of a face is fixed to zero, fixed to one, or free, so there are 3^d faces overall. A face meets K if and only if it is not entirely contained in K^c. Counting faces meeting K according to their fixed coordinate set gives $\sum_{U \subseteq [d]} |\pi_U K|$. The missing-class correction counts the coordinate subsets not shattered by K, which is $2^d - |\mathrm{Shat}(K)|$. Adding gives the identity. □

Remark 5.2 (Relation to cube-face inequalities)**.** *Bollobás–Radcliffe's face-isoperimetric theorem [1] characterizes extremal contained-face counts for fixed-size Boolean sets. The decomposition above separates that face statistic from the number of shattered coordinate supports. Our protected-family optimization requires both terms to be evaluated under the marginal-surjectivity condition.*

6 The protected-set objective as a polymatroid rank

The exact index has a further structural expression. Although the matroid tools used below are classical, the decomposition records an additional property of this particular readout cost without relying on any fixed-cardinality case. For the empty protected set we extend the index by its all-generic Moore quotient, which has one state for every support, namely $I_d(\varnothing) = 2^d$.

Theorem 6.1 (Truncated-partition-matroid decomposition)**.** *For every $K \subseteq \{0,1\}^d$, including the empty set, set*

$$R_d(K) = \sum_{U \subseteq [d]} \min\bigl\{|\pi_U K|,\, 2^{|U|} - 1\bigr\}.$$

Then $I_d(K) = 2^d + R_d(K)$. Moreover R_d is a normalized, nondecreasing, integer-valued submodular function on the ground set $\{0,1\}^d$:

$$R_d(A) + R_d(B) \geq R_d(A \cup B) + R_d(A \cap B) \qquad (A, B \subseteq \{0,1\}^d).$$

More precisely, R_d is a sum of ranks of truncated partition matroids.

Proof. For integers $0 \leq m \leq n$ we have $m + \mathbf{1}_{\{m<n\}} = 1 + \min\{m, n-1\}$. Apply this to $m = |\pi_U K|$ and $n = 2^{|U|}$, and sum over all 2^d supports. For fixed U, partition the ground set $\{0,1\}^d$ into the fibers of π_U. The partition matroid which permits at most one representative from each fiber has rank $|\pi_U K|$ on K. Truncating this matroid to rank $2^{|U|} - 1$ gives the summand of R_d. Matroid ranks are normalized, nondecreasing and submodular, as are their sums. □

Proposition 6.2 (Exact diminishing returns and the final-label plateau)**.** *For $x \notin K$, the additional port cost is exactly*

$$I_d(K \cup \{x\}) - I_d(K) = \#\bigl\{U \subseteq [d] : \pi_U(x) \notin \pi_U K, \quad |\pi_U K| \leq 2^{|U|} - 2\bigr\}.$$

Consequently, if K is nonempty the increment is at most $2^d - d - 1$, and this upper bound is sharp for $K = \{0^d\}$ and $x = 1^d$. Every addition with $|K| \leq 2^d - 2$ has strictly positive cost, but the very last full word has cost zero: $I_d(\{0,1\}^d \setminus \{x\}) = I_d(\{0,1\}^d) = 3^d$.

Proof. In each rank summand of Theorem 6.1, adding x increases the number of projection classes precisely when $\pi_U(x) \notin \pi_U K$, and this increases the truncated rank by one precisely when the old rank is below $2^{|U|} - 1$. For nonempty K, supports of size at most one contribute no increment; the remaining number is $2^d - d - 1$, and the specified complementary singleton construction gains one on every such support. If $|K| \leq 2^d - 2$, the full support $U = [d]$ satisfies both requirements, giving an increment of at least one. If K consists of all but one complete word, every proper-support projection is already full, and the full-support rank is already at its cap. Thus the final addition costs zero. The complete source has precisely 3^d port classes, proving the final equality. □

Remark 6.3 (Submodularity and constrained extremals)**.** *The matroid truncation and rank submodularity used here are standard [5, 6]. They supply a structural description of the port objective. The fixed-cardinality problem with marginal surjectivity is a constrained extremal problem, and it is not solved by an unconstrained submodular-minimization routine.*

7 Concluding remarks

The minimum autonomous-port state count is governed by the complete family of protected coordinate projections, together with the supports on which some complete assignments remain unprotected. The sharp initial thresholds, face-vector separation, and polymatroid structure describe complementary features of this objective. The formulas distinguish the separately metered readout interface from the complete semantic source retained by its host.

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